

# Jihun Rho

GPU Performance Engineer

Computer Vision Engineer

B.S. Electronic Engineering, Sejong University

*CUDA · Computer Vision · Real-Time Systems*

## CONTACT


Seoul, South Korea

+82 10-2292-2510

hunjirho@gmail.com

GitHub profile

Portfolio website


## CORE SKILLS

**CUDA / GPU**

C++, CUDA kernels, memory tuning, stream overlap, occupancy

**Profiling**

Nsight Systems, Nsight Compute, performance analysis

**Computer Vision**

OpenCV, image processing, detection, tracking, calibration

**Inference**

TensorRT, YOLO, camera-radar fusion, real-time evaluation

**Systems**

C/C++, Multithreading, Linux, Windows

**Robotics**

Robot control, REST, TCP/IP, Modbus, Digital I/O

# PROFILE

GPU Performance Engineer | Computer Vision Engineer

GPU Performance Engineer with 4 years and 8 months of hands-on CUDA/C++ experience, specializing in GPU kernel optimization, high-performance image processing, and industrial machine vision. Bringing 7 years and 6 months of combined industry and research experience spanning computer vision, deep learning, and robotics.

**Portfolio:** 931njhthe-star.github.io

# EXPERIENCE

### Research Engineer — CUDA & GPU Performance Optimization

Enclony — Advanced Technology Research Institute | Mar 2021 – Oct 2025

- Developed and optimized high-performance CUDA/C++ machine-vision software for industrial pharmaceutical inspection systems.
- Redesigned image-processing pipelines with selective CUDA acceleration for compute-intensive workloads, retaining efficient CPU processing for lighter tasks and optimizing CPU–GPU workload distribution.
- Achieved 11.2× higher image-processing throughput than a single-core CPU baseline and approximately 40% higher throughput than a multi-core CPU-only implementation, reaching 2.52 million images per hour.
- Processed 10 RGB images (640×480) and 2 3D camera-derived height maps per object for multi-view surface inspection.

Technical Skills: C++, CUDA, CPU-GPU Pipeline Optimization, Computer Vision

### Researcher — Deep Learning & Computer Vision

Sejong University — Intelligent Vehicle Recognition Systems Laboratory — Nov 2019 – Oct 2020

- Vehicle Detection: Evaluated YOLOv3/v4 and MobileNet-SSD on rooftop-captured video footage. Curated tens to hundreds of thousands of extracted frames, managed JSON/text annotations, and performed detection evaluation and error analysis.
- Traffic Sign Recognition: Developed a 3D-to-2D projection simulation using camera geometry and coordinate transformations. Evaluated LeNet and SVM recognition performance for traffic signs rotated around different 3D axes, analyzing the effect of rotation angles on accuracy.

Technical Skills: C++, Python, 2D/3D Computer Vision, YOLO, MobileNet-SSD, LeNet, SVM

### Software Engineer — Robotics & Vision Systems

INA Oriental Motor — Technology Research Center | Jan 2018 – Oct 2019

- Planned and integrated an industrial robotics demonstration system combining autonomous mobile robots, six-axis robots, 3D vision systems, and pneumatic grippers.
- Developed integration software using Python, REST APIs, TCP/IP, Modbus, and digital I/O to connect heterogeneous devices and their existing control systems.
- Implemented monitoring, motion coordination, and gripper-control workflows for multi-device operation.
- Integrated deep-learning-based 3D object recognition and random bin-picking solutions into robotics demonstrations.

Technical Skills: Python, Industrial Robot and Sensor System Integration, REST API, TCP/IP, Modbus, Digital I/O

## PROFESSIONAL DEVELOPMENT

**Multi-Agent AI Orchestration Program**

SK Networks Family × Encore AI Campus | Aug 2026 – Jan 2027 (Expected)

**Machine Learning Practical Course**

Goot Academy | Dec 2018 – Jan 2019

**Python Web Programming**

Korea Software HR Development Institute | Jul – Aug 2018

**Arduino Electronics Workshop**

E2 Electronics Academy | Jun – Jul 2018

**Security & Network Administrator Program**

Korea Information Security Academy | Sep 2016 – Jan 2017

## CERTIFICATIONS

Information and Communication Engineer

Radio Operator Engineer

Industrial Engineer — Electronics

Industrial Engineer — Computer Maintenance Control

PCCE — Python (2026)

CCNA (Cisco Certified Network Associate) — Expired

Computer Specialist in Spreadsheet & Database Level 2

ATC Level 2

Class 1 Ordinary Driver's License

# SELECTED PROJECTS

Applied work across perception, robotics integration, and AI systems.

### DriveFusion — Real-Time Camera–Radar Perception & Evaluation

**View project page**

Independently built a CARLA-based system combining YOLO11n inference through a C++/CUDA TensorRT module, vehicle tracking, radar-assisted distance estimation, and a Qt monitoring dashboard. Across nine weather and time-of-day test conditions (16,491 common samples), radar-assisted tracking reduced overall distance MAE by 5.41%; the reduction reached 39.42% on 437 radar-corrected samples.

Technologies: C++, CUDA, TensorRT, YOLO11n, Qt, CARLA, radar fusion

### Robotics & 3D Bin-Picking System Integration

**View project page**

Integrated a MiR AMR, Doosan six-axis robot, Solomon 3D vision, and pneumatic gripper control into a coordinated bin-picking system. Connected device controllers through REST, TCP/IP, Modbus, and digital I/O.

Technologies: Robot control, 3D vision, REST API, TCP/IP, Modbus, digital I/O

### SubSync — YouTube English-Learning Extension

**View GitHub repository**

Project manager for a five-person team building a Chrome Extension MVP with synchronized English/Korean subtitles, vocabulary saving, and an AI tutor. The backend test suite passed all 79 tests.

Technologies: Chrome Extension MV3, FastAPI, AI agent, Supabase, Redis, DB

## EDUCATION

B.S. in Electronic Engineering — Sejong University | Mar 2009 – Feb 2018

GPA: 3.38 / 4.5 | ABEEK-accredited engineering program

## SELECTED AWARDS

**Excellence Award — 2nd ICT CoC Hackathon (2020) Ministry & Seoul Metropolitan Government Level**

Awarded by the National IT Industry Promotion Agency (NIPA) and the Seoul Metropolitan Government for YOLOv3 aerial detection of unauthorized buildings and rooftop gardens.

Technologies: C++, YOLOv3, deep learning, computer vision, object detection

**Bronze Award — Sejong University Creative Design Competition (2017) | University-Level**

Developed an optical-sensor digital ruler application for distance measurement.

Technologies: Optical sensors, sensor-based measurement, software development

**4th Place — Sejong University Software Competition (2016) | University-Level**

Placed 4th in a university algorithm-programming competition; focus on algorithm design, problem solving, and C programming.

**Outstanding Talent Award — 2016 Winter Career Competency Enhancement Camp | JobKorea & Sejong University**

JobKorea recognized my outstanding participation and presentation at its winter employment camp with Sejong University.

**HIGH-SCHOOL RECOGNITION | Seongho High School**

Physics Science Competition | May 2008

Top Award — Essay Writing Assessment Competition, Mathematics & Science Division | Oct 2007